

RUDRABHA CHAKRABORTY

AI Engineer | Generative AI, Agentic Systems & LLM Evaluation

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PROFESSIONAL SUMMARY

AI Engineer with 1.5+ years building production Generative AI for commercial insurance underwriting and claims on Microsoft Azure. Ships agentic RAG assistants, retrieval augmented document extraction, and guideline to rules engines, and treats evaluation as part of the product: regression harnesses, ground truth scoring, and DSPy prompt optimization. Core engineer on an extraction platform running at **95.4% mean field accuracy** across 362 live submissions. M.Sc. Computer Science, NIT Trichy.

TECHNICAL SKILLS

LLMs & Agents: GPT 5.x, GPT 4.1, Claude, Agentic RAG, Planner Executor Agents, Tool Calling, Tool RAG, Model Context Protocol (MCP), LangChain, LangGraph, DSPy
LLM Engineering: Context Engineering, Prompt Decomposition and Chaining, Structured Outputs, LLM Guardrails, Hallucination Mitigation, Intelligent Document Processing
Retrieval & Data: Hybrid Search (BM25 and dense embeddings), ChromaDB, Qdrant, Azure Document Intelligence, Mistral OCR, PyMuPDF, pdfplumber, pandas
Evaluation & LLMops: Eval Driven Development, Regression Test Harnesses, Ground Truth Adjudication, Model and OCR A/B Benchmarking, OpenTelemetry
Backend & Cloud: Python, SQL, FastAPI, Celery, Redis, MongoDB, Azure (OpenAI, Container Apps, Functions, Cosmos DB, Service Bus), Docker, Azure DevOps CI/CD

WORK EXPERIENCE

AI Engineer | BluePond AI Jul 2025 to Present

LossRuns | Loss Run Extraction Engine *Azure OpenAI, Hybrid Search, FastAPI, Celery, Redis*

- Turned carrier loss runs with no shared format into structured claims data by building a retrieval augmented pipeline with hybrid BM25 and dense search and **55+ carrier specific prompt routes**, reaching **95.4% mean field accuracy** across 15 fields on 362 live submissions, 54 fully zero touch.
- Stopped LLM cost from scaling with volume by designing a hybrid line of business classifier: rules and fuzzy matching settle clear claims, and only ambiguous ones go to a batched LLM fallback.
- Slimmed the service by moving dense retrieval from local sentence transformers to OpenAI embeddings, removing PyTorch from the image; A/B benchmarked Mistral OCR vs Azure Document Intelligence to drive the OCR vendor choice.

CLAIMS | Claims Document Intelligence Platform *DAG Orchestration, Cosmos DB, Service Bus, Azure Document Intelligence*

- Solved mixed claim bundles (adjuster reports, estimates, invoices in one PDF) with a multi label page classifier that splits each file into typed sections before extraction.
- Cut tenant onboarding from a code release to a config change by extending the DAG orchestration engine with invoice and estimate pipelines and making loss run, property schedule, and ACORD form flows config driven.
- Replaced duplicated loss cause logic with one multi tenant LLM classification microservice using a global, app, and tenant config hierarchy, now serving both Claims and Loss Runs.

SENSE | AI Underwriting Copilot *Agentic RAG, GPT 5.2, ChromaDB, Qdrant, FastAPI*

- Built the planner executor core of an agentic underwriting assistant: Tool RAG over ChromaDB selects tools, the LLM planner emits a structured tool call plan, and response composers ground every answer in retrieved submission data.
- Caught conflicting rules before release with LLM contradiction detection over rules extracted from guideline PDFs, backed by versioned rule diffing and governance APIs.
- Kept generated insights honest with an XML structured prompt that reasons only over rule engine verdicts, so every insight traces back to a specific rule.

AI Engineering Intern | BluePond AI Jan 2025 to Jun 2025

Broker Slip Extraction Pipeline (End to End) *DSPy, Prompt Chaining, GPT 5.2, Celery, Azure*

- Raised exact match accuracy from **72.3% to 79.9%** on a 57 case regression set by decomposing a 19 field prompt carrying 56 negative instructions into focused coverage and peril prompts and removing hallucinated flood and earthquake limits.
- Replaced manual prompt tuning with **DSPy** prompt optimization, using the regression set as the evaluation metric, so prompt changes became measurable and repeatable.
- Built an Azure Functions evaluation service that scores model output against human reviewed ground truth daily and emails a weekly accuracy report, plus regression harnesses that gate prompt and model changes across three pipelines.

ACADEMIC PROJECTS

Empathy Aware Conversational AI *LLaMA 2, RAG, Fine Tuning*

- Built a RAG chatbot on LLaMA 2 fine tuned on EmotionLines, improving contextually relevant responses by 20% and empathetic response accuracy by 30%.

Retail Sales Forecasting *Python, scikit learn, Random Forest*

- Built a Random Forest forecaster on Walmart sales data (WMAE 1528.71 in validation), outperforming KNN and Extra Trees baselines with K Fold cross validation.

EDUCATION

Year	Degree / Examination	Institution / Board	CGPA / Percentage
2025	M.Sc. Computer Science	National Institute of Technology (NIT), Tiruchirappalli	8.15 / 10
2023	B.Sc. (Hons) Statistics	St. Xavier's College, Kolkata	7.62 / 10
2020	Class XII (WBCHSE)	St. Lawrence High School, Kolkata	96.00%
2018	Class X (WBBSE)	St. Lawrence High School, Kolkata	91.00%

Achievements: DST INSPIRE Scholarship, Government of India • Top 1% in WBCHSE Class XII examinations, 2020